

IN SEARCH OF CALABI-YAU 3-FOLDS

Moduli Space of n_3 Configurations

Henry Morin

Under the Supervision of Jenia Tevelev

July 24, 2026



SECTION 1

Calabi-Yau Manifolds

I Definition

- ◉ "Nice" subsets of $\mathbb{R}^N$
- ◉ Locally flat, that is locally $\mathbb{R}^d$
- ◉ Classical object of study in physics and mathematics

Examples

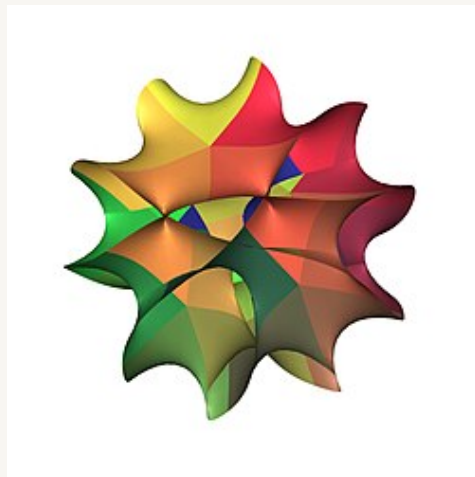
- ◉ Open subsets of $\mathbb{R}^N$
- ◉ Spheres
- ◉ Tori
- ◉ Projective Space ($\mathbb{P}^n$)

I Definition

- ◉ Globally "flat" manifolds M
- ◉ Even dimensional $2d$
- ◉ Trivial Canonical Line Bundle $\omega_M = \bigwedge^{2d} T^*M = 0$

Examples

- ⊙ Elliptic curves (complex tori)
- ⊙ Smooth 4-dimensional quartic surfaces
- ⊙ Smooth 6-dimensional quintic surfaces



■ Why do we Care?

- ⊙ CY manifolds naturally arise in physics
- ⊙ Classification of CY manifolds important in string theory
- ⊙ CY manifolds are interesting in their own right

I General Method

- ◉ Degree $n + 1$ polynomial equation in n variables f
- ◉ Take $Z(f) := \{z \in \mathbb{C}^n \mid f(z) = 0\}$
- ◉ Add points at ∞
- ◉ Desingularize

■ This is Hard

- ⊙ Very difficult to obtain f of higher degree
- ⊙ Once you have f , desingularization can be expensive
- ⊙ "Shape Shift" - Admit many models

SECTION 2

n_3 Configurations

I Definition

- ◉ "Points" $P = \{1, \dots, n\}$
- ◉ "Lines" $L \subset 2^P$
- ◉ n_3 condition: $|\ell| = 3$ and $|\{\ell \mid i \in \ell\}| = 3$

n	n_3 configurations
7	1
8	1
9	3
10	10
11	31
12	229
13	2036

I Definition

- ⊙ Fix n_3 configuration (P, L)
- ⊙ A **moduli space** is the set M of solutions to a polynomial
- ⊙ Such that $z \in M$ corresponds to one way of drawing (P, L)

Setup

- ◉ M is given by the set of points $(p_1, p_2, \dots, p_n) \in (\mathbb{P}^2)^n$
- ◉ Such that p_i, p_j, p_k are collinear in $\mathbb{P}^2$ iff $i, j, k \in \ell$ for some $\ell \in L$.
- ◉ Given by n equations of multi-degrees $(0, \dots, 1, 1, 1, \dots, 0)$
- ◉ Successively applying the restrictions we obtain

$$M \subset M_{n-1} \subset M_{n-2} \subset \dots \subset M_1 \subset (\mathbb{P}^2)^n$$

- ◉ Quotient by $\mathrm{PGL}(3)$

Computing ω_M

- ⊙ Canonical class ω of a hypersurface is given as $\iota^*(\omega \otimes \mathcal{O}(d))$
- ⊙ $\omega_{M_1} = \iota^*(\omega_{(\mathbb{P}^2)^n} \otimes \mathcal{O}((1, 1, 1, 0, \dots, 0)))$

$$\omega_{M_1} = \iota^*(\omega_{(\mathbb{P}^2)^n} \otimes \mathcal{O}((1, 1, 1, 0, \dots, 0)))$$

$$\omega_{M_2} = \iota^*(\omega_{M_1} \otimes \mathcal{O}((1, 0, 1, 1, 0, \dots, 0)))$$

$$\vdots$$
$$\vdots$$

$$\omega_M = \iota^*(\omega_{M_{n-1}} \otimes ((0, 0, \dots, 0, 1, 1, 1)))$$

■ M is Calabi-Yau!

- ⊙ Since $\omega_{(\mathbb{P}^2)^n} = \mathcal{O}(-3, -3, \dots, -3)$
- ⊙ This becomes (by omitting $\mathcal{O}$):

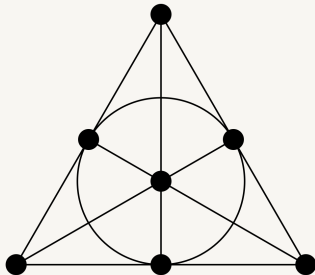
$$\begin{aligned}(-2, -2, -2, -3, \dots, -3) &= \iota^*((-3, -3, \dots, -3) \otimes (1, 1, 1, 0, \dots, 0)) \\(-1, -2, -1, -2, \dots, -3) &= \iota^*((-2, -2, -2, -3, \dots, -3) \otimes (1, 0, 1, 1, 0, \dots, 0)) \\&\vdots \\(0, 0, \dots, 0) &= \iota^*((0, 0, \dots, 0, -1, -1, -1) \otimes (0, 0, \dots, 0, 1, 1, 1))\end{aligned}$$

So $\omega_M = 0$!

$$n = 7$$

| Summary

- ◉ Number of configurations: 1 (Fano Plane)
- ◉ Moduli Space: $f = 1$
- ◉ Not realizable in $\text{char} \neq 2$

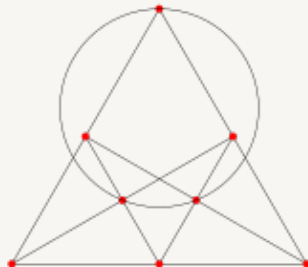


$$n = 8$$

Summary

- ◉ Number of configurations: 1 (Möbius Cantor Configuration)
- ◉ Moduli Space: $f = x^2 - x + 1$
- ◉ 2 realizations corresponding to

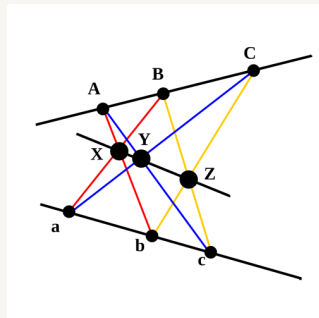
$$x = \frac{1 \pm \sqrt{-3}}{2}$$



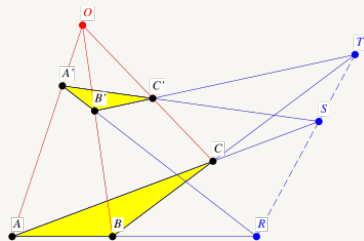
$$n = 9$$

Summary

- ⊙ Number of configurations: 3
- ⊙ Moduli Spaces: $f = 0$
- ⊙ One "wrong" dimension (Pappus Configuration)



- ◉ Number of configurations: 10
- ◉ One not realizable in any characteristic
- ◉ Moduli Spaces: Four are $K3$ surfaces, five are given by $f = 0$
- ◉ One "wrong" dimension (Desargues Configuration)



I Summary

- ⊙ Number of configurations: 31
- ⊙ 7 known to be given by $f = 0$, 12 quintic, 5 sextics, 3 septics, 4 octics
- ⊙ We currently are working towards desingularization of quintic examples
- ⊙ We have three (non confirmed) Calabi-Yau matches

I Ongoing Work

- ⊙ Analyse double cover models by blowups of $\mathbb{P}^3$
- ⊙ Utilize $K3$ fibration structure on M to obtain smooth model
- ⊙ Utilize elliptic fibration structure on M to obtain smooth model
- ⊙ Study rigidity of M
- ⊙ Study modularity of M

- ⦿ E. Sink, Line Configurations and K3 Surfaces. <https://arxiv.org/abs/2312.07542>
- ⦿ C. Myer, Modular Calabi-Yau Threefolds.
<https://www.ams.org/books/fim/022/fim022-endmatter.pdf>